

BILAL ZAHID

AI Infrastructure Engineer | Agentic Systems & LLM Evaluation

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SUMMARY

AI infrastructure engineer and technical lead with 6+ years building production systems end-to-end. Architected Google's Gemini training and evaluation infrastructure at Turing, agentic simulation over 77 real-world APIs, synthetic-world generation, and RLHF/SFT across Gemini, Grok, and ServiceNow AI. Ships distributed, cloud-native platforms across enterprise SaaS and healthcare while leading engineering teams and shaping technical direction.

PROFESSIONAL EXPERIENCE

Google Services - LLM Python Developer



Jun 2024 - Present

- **Architected the Gemini Gym's agentic simulation environment (Generalized Agent APIs):** rebuilt 77 real-world APIs as a deterministic, resettable sandbox with 3,200+ automated tests, so Gemini trains and is graded on multi-turn tool use without touching live data.
- **Developed a multi-stage function-calling validation framework** across 75+ tool schemas, enforcing schema integrity, cross-provider compatibility (Google, OpenAI, Anthropic), and signature consistency, catching defects Google's own GenAI SDK missed and lifting quality coverage to roughly 90%.
- **Built a sim-to-real fidelity and regression suite** across 36+ services, proving simulated tools matched production behavior, cutting downstream defects 20%, and wiring validation directly into CI.
- **Engineered DBGen, the Gym's synthetic-world generation pipeline**, end-to-end across generation, evaluation, scaling, and quality: turns a one-paragraph spec into a fully-populated, schema-accurate training world with an auto-generated evaluation rubric, validated across three generations of Gemini models.
- **Designed the Gym's robustness-evaluation harness (Benchmark Mutation Suite):** degrades tool schemas, input modality, and call format across 10 industry benchmarks (including the 109-task Toolathlon and 231-task MCP Universe), benchmarking Gemini against GPT-5 and Claude under real-world failure conditions.
- **Drove human ground-truth evaluation of AI coding agents (SWE Trajectory Evaluation):** graded model trajectories against a 15-tag quality rubric at trace and step level, producing the answer key used to measure how reliably models judge their own and each other's work.
- **Spearheaded supervised fine-tuning (SFT) and RLHF data curation** for Gemini, Claude, Grok, and ServiceNow AI, producing code-centric training datasets that raised contextual accuracy 18% and cut inference latency 45 ms.
- **Led a 6-person team of AI trainers and engineers** across dataset curation, infrastructure, and model evaluation, setting sprint priorities and reviewing code to keep delivery on schedule.

Python · React · Next.js · Django · FastAPI · Pydantic · TensorFlow · PyTorch · Hugging Face · AWS · Docker · SQL · Pinecone · Prompt Engineering · RLHF/SFT

Senior Full-Stack Engineer

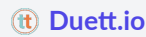


Sep 2022 - Sep 2024

- **AI-Powered Revenue Operations (GTM):** orchestrated autonomous multi-agent CRM workflows on a serverless, event-driven AWS Lambda architecture with NLP-driven pipelines and React/Next.js analytics dashboards, lifting sales-team productivity 30%, operational efficiency 250%, and close rate 150%.
- **Global Tax Compliance Platform:** founding engineer on the world's first AI-powered indirect-tax platform (sales tax, VAT, GST), architecting registration, calculation, and filing automation with geolocation-based jurisdiction detection, Stripe / QuickBooks / HMRC integrations, and Kubernetes workers and cron jobs on AWS EKS for high-volume sync and scheduled compliance.
- **AI Data & Connector Automation:** created a documentation-to-connector pipeline that crawls API docs, embeds them with OpenAI and Pinecone, and generates low-code connector manifests on AWS Lambda, proven across 7+ third-party APIs, plus natural-language-to-SQL tooling that turns plain questions into validated queries.
- **EdTech Learning Platform:** delivered an earlier live learning platform (Ruby on Rails, React/Next.js) with real-time class tools and a companion Zoom note-taking app, lifting user engagement 40%.
- **Team leadership (across tenure):** owned system reliability and engineering quality for a 7-person team, tuned PostgreSQL, Pinecone, and MongoDB (bugs down 20%, maintainability up 15%), drove technical roadmap discussions with the CEO and CTO, and mentored 5 engineers through onboarding and delivery.

Ruby on Rails · Python (FastAPI) · React · Next.js · TypeScript · OpenAI · AWS (Lambda, RDS, EC2, S3, CloudWatch, API Gateway, SES, CDN) · Salesforce · HubSpot · PostgreSQL · MongoDB · Pinecone · Redis · Docker · Stripe · Fivetran · PropelAuth

Senior Software Engineer



Jan 2021 - Aug 2022

- **Architected a HIPAA-compliant, cloud-native healthcare platform (DUETT)** solving a core care-coordination and interoperability problem, connecting care coordinators with home-health providers through a resilient backend and reshaping care delivery across Central Indiana.
- **Created a provider-matching engine** on Django and Flask that translated patient needs into optimized care-coordinator assignments, backed by rigorous safeguards for protected health information.
- **Shipped scheduling and workflow-optimization features** that lifted sales **15%** and cut bounce rate **25%** on a React and Django stack.
- **Integrated predictive-analytics models**, developed with data science partners, into the provider-matching pipeline, raising match accuracy **20%**.
- **Scaled backend services on AWS** to sustain **10,000 requests per minute** while cutting API latency **25%**, keeping the platform resilient through provider-network growth.
- **Expanded Duett's provider network** by integrating case managers and clients into the platform, establishing Duett as a recognized leader in healthcare-service matching.

Python (Django, Flask) · React · PostgreSQL · MongoDB · AWS (EC2, S3, SES) · Jenkins · Grafana · MUI

Software Engineer



Mar 2020 - Jan 2021

- **Built a digital health platform ("Reflect")** powered by an intelligent recommendation algorithm that matched patients to therapists based on specific stressors, lifting match accuracy **30%**.
- **Developed a calendar-orchestration microservice** on Django handling asynchronous scheduling for therapist availability, eliminating scheduling conflicts **40%** and shipping as a standalone, reusable service.
- **Engineered secure, scalable microservices** and RESTful APIs on AWS within a cross-functional Agile team, sustaining **10,000 requests per minute** while cutting data-retrieval time **25%**, with privacy safeguards built into every service handling sensitive health data.

Python (Django) · React · AWS (EC2, Lambda, RDS, CloudWatch, CDN) · MongoDB · PostgreSQL · MUI · Mailgun

TECHNICAL EXPERTISE

AI / LLM Engineering

Agentic AI, Tool & Function Calling, Function Schema Validation, Multi-Turn Reasoning, Prompt Engineering, LLM Orchestration

LLM Evaluation & Data

LLM Evaluation, Evaluation Pipelines, Benchmark Harnesses, Robustness Testing, Simulation Environments, Synthetic Data Generation, Model Evaluation, RLHF/SFT

ML Frameworks

PyTorch, TensorFlow, Hugging Face, NLP

Cloud & DevOps

AWS, GCP, Docker, Kubernetes, CI/CD, CircleCI, Jenkins, Datadog, Grafana, Observability, Linux, Git/GitHub, Agile/Scrum

Languages

Python, TypeScript, JavaScript, SQL, Ruby, GraphQL, HTML5, CSS3

Backend & APIs

FastAPI, Django, Flask, Ruby on Rails, REST, GraphQL, Pydantic

Architecture & Design

Microservices, Event-Driven Architecture, System Design, Distributed Systems, API Design, Performance Optimization, HIPAA Compliance, OWASP

Frontend

React, Next.js, TypeScript, Material UI, Bootstrap, Jest

Data & Stores

PostgreSQL, MongoDB, MySQL, Pinecone, Redis, Elasticsearch

EDUCATION

FAST-NUCES (National University of Computer & Emerging Sciences)

Bachelor of Science in Computer Science

Lahore, Pakistan

Aug 2016 - May 2020